

aTOMos v3.6.1.13

Physics foundation for application kernels

Conservation, constitutive closure and measurement
Exact operators with explicit physical meaning

Mechanics and thermodynamics · electromagnetism
Calibration and uncertainty · exact discrete energy balance

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A focused addition of physical laws and model closures that directly strengthen the existing aTOMos interfaces. The new finite mechanical profile has a derived discrete energy balance. Optical loss, material storage, thermal coupling and measurement uncertainty retain explicit equations and domains.

The parent corpus is preserved and referenced without repetition. Mathematical formalization; no new runtime or physical performance claim. Prepared with AI assistance.

Scope, lineage and method

This supplement addresses a specific requirement: strengthen the foundation with actual physics and only the additions that help its formalization. It adds a minimal law-profile contract, finite mechanical power balance, thermodynamic and electromagnetic closure, and measurement/output conditions. It does not repeat the complete 145-page R12 corpus or introduce unrelated application families.

The parent is `atomOS_3_6_1_12_PHOTONIC_INTERFACE`, committed at 96a2192. Its complete PDF and source identities are bound in `source/PARENT_BINDING.json`. Its XOP word/operator contracts, WANTWOMBAN equations, ASA/NA and JK transitions, orbital models and receiver profiles remain the authority for those definitions. R13 uses named templates over that vocabulary; no existing profile is silently replaced.

Established physical laws retain their external scientific provenance. Constitutive examples state the selected material assumptions. The new integration decisions and finite-step proofs are derived in this document. Mathematical consistency, faithful implementation and physical adequacy are different claims with different evidence.

Following the continuing work method, no reference programs, numerical examples, algorithm tests, simulations, benchmarks or physical experiments were executed. The work consists of equation derivation, primary-source inspection, critical review and document preparation. The proofs are readable conditional arguments, not proof-assistant certificates. Rendering and file hashes verify delivery and provenance, not the proposed system's performance.

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1 A minimal physical-law layer for application kernels

R13 strengthens aTOMos where a physical law supplies a missing relationship, constrains a proposed coupling or justifies a useful implementation choice. It is a focused supplement to R12. The existing XOP representation, literal ASA/NA and JK rules, WANTWOMBAN packet, orbital and receiver equations remain in their bound parent. Repeating that corpus would not strengthen these laws.

The new profile is XOP-PHYSICS-R1. Its mathematical bodies use the existing XOP-R1 operators; it does not allocate new opcodes or claim a new executable backend. A physical specialization imports only the laws and closures needed by its declared interface. A purely digital permutation does not need a continuum electromagnetic model merely because real hardware will eventually execute it.

1.1 Five distinct parts of a physical specialization

Part	What it supplies	Example in the existing model
Balance law	Conserved quantity, flux or production relation	Charge continuity; energy and momentum balance
Constitutive model	Material or component response under stated conditions	Polarization dynamics; thermoelastic free energy
Reduction	A finite or simplified model with its scope	Fixed modal basis; scalar optical field; lumped temperature
Discrete evolution	A selected finite state transition	Existing backward Euler; new energy-balanced midpoint
Observation	How physical state produces recorded evidence	Acquisition window, calibration, quantization and uncertainty

Conservation equations alone do not specify a material. A constitutive law alone does not provide boundary data. Exact arithmetic alone does not select a conservative time update. A recorded count alone does not fix a physical measurand. Keeping these parts explicit allows specialization without hidden changes of meaning.

1.2 Normative profile contract

A selected physical profile declares the finite record

$$\Pi = (\mathcal{T}, \mathcal{X}, \mathcal{U}, \mathcal{Y}, \Theta, \mathcal{B}, \mathcal{C}, \mathcal{A}, \mathcal{N}, \mathcal{O}, \mathcal{E}). \quad (1)$$

$\mathcal{T}$ contains units, frames and time conventions. $\mathcal{X}$ declares state and initial conditions; $\mathcal{U}, \mathcal{Y}$ declare inputs and outputs. Θ contains parameters with source identities and joint uncertainty. $\mathcal{B}$ lists applicable balance laws. $\mathcal{C}$ supplies constitutive closures. $\mathcal{A}$ lists approximation and admissibility hypotheses. $\mathcal{N}$ identifies a finite representation and the chosen update. $\mathcal{O}$ defines acquisition and calibrated observations. $\mathcal{E}$ records the claimed output bounds and evidence status.

Each law entry has a qualified name/version, argument types, equation body, domain, boundary terms, source and a status. Status distinguishes a definition, a derived identity, an adopted physical law, a constitutive assumption and a measured parameter. The same source tag cannot turn one status into another. The editable normative field contract is `formal/PHYSICS_PROFILE.md`. This is a record definition, not a claim that a new parser implements it.

For a finite state model, the mathematical content can be written

$$\dot{x} = f_{\Pi}(t, x, u; \Theta), \quad y = g_{\Pi}(t, x, u; \Theta), \quad B_j(t, x, \dot{x}, u; \Theta) = 0, \quad (2)$$

with declared inequalities and initial/boundary conditions. The B_j are relations to derive or impose consistently, not extra independent equations to append blindly to a fully determined system. Redundant balances are identified; inconsistent initial data or incompatible closures do not define an admitted solution. A finite ODE must meet the existing existence and uniqueness domain. Algebraic constraints need elimination or a separately defined constrained formulation; they are not silently handled by ODE.

1.3 Required connections to the existing seed

Existing quantity	Connection required by R13
$t_\lambda, H_{z,\lambda}$	Chosen electromagnetic transfer, material response, boundary data and power normalization
$P_{\text{abs}}, P_{\text{heat}}$	A shared energy ledger distinguishes absorption, stored excitation and dissipated heat
M, C, K and strain	Fixed geometry/basis or declared motion, force/work convention and constitutive response
T, ζ, a	Thermal/material energy stores and rates with the relevant temperature and coupling domains
ASA/NA, JK and actuator command	The word selects an action; a separate actuator equation supplies force, voltage, light or heat
WI measurement and calibration	Acquisition history, physical units, clock/frame binding, uncertainty and measurand equation
Exact operator graph	Captured physical inputs and hypotheses, complete finite operands, explicit unresolved outcomes

For example, a selector $b \in \{0, 1\}$ may choose $K(b) = K_0 + b(K_1 - K_0)$. It does not by itself specify the work required to switch stiffness. For an unchanged coordinate q , that storage change is $q^T(K_1 - K_0)q/2$ and must be attributed to the switching interaction. Signed damping or stiffness remains mathematically expressible; a passive energy theorem applies only when its sign hypotheses hold. An active element needs its actual supply and constitutive model.

1.4 Representation preserves laws only through an explicit map

Let $x = \mathcal{R}(z)$ reconstruct a finite physical model from encoded state z , and let F_h be a chosen digital transition. A continuous-law identity does not automatically give

$$\mathcal{B}_{j,h}(\mathcal{R}(z), \mathcal{R}(F_h(z, u)), u) = 0. \quad (3)$$

Here $\mathcal{B}_{j,h}$ is a separately defined discrete balance, rather than the continuous residual B_j with its arguments silently changed. The discrete relation has to be derived for the selected step. The law may instead provide an error bound or a nonzero residual ledger. Likewise, a finite basis is a declared approximation to a continuum field: exact evaluation of its coefficients and trajectory does not erase omitted modes. The electromagnetic and mechanical chapters state the necessary finite-representation and integral domains.

One-bit word packing preserves the encoded operators, parameters and inputs. It does not turn a finite word mask into an electromagnetic absorber, identify a code bit with a photon, or establish that a material follows a named law. Those are explicit realization mappings. The R12 photonic refinement contract continues to govern any proposed hardware executor.

2 An exact finite step with a derived mechanical energy balance

This addition supplies a specific useful kernel specialization. It removes algorithmic energy loss from the fixed linear model's discrete balance while retaining physical damping. It does not replace the existing CGK or orbital profiles silently and does not assert exact continuous-time trajectories.

2.1 Selected model and units

Choose a finite number of length-valued coordinates q in a fixed declared modal basis, velocity v , and a fixed reference configuration. Let

$$\dot{q} = v, \quad M\dot{v} = \bar{f} - Kq - Cv, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad C = C^T \geq 0. \quad (4)$$

Here the displayed $\bar{f}$ denotes the force selected for the interval's discrete supply; a held-force physical model may use the same constant value. The quantities M, C, K have units kg, kg/s and kg/s² when all q coordinates have length units. Generalized coordinates of mixed dimensions require the inherited tuple/block rules, not a false uniformly dimensioned matrix. The storage is

$$E(q, v) = \frac{1}{2}v^T Mv + \frac{1}{2}q^T Kq, \quad \dot{E} = v^T \bar{f} - v^T Cv. \quad (5)$$

The continuous identity follows by differentiating the two quadratic terms and substituting the model. Its damping term becomes heat or another named output store in a coupled physical model, as derived in the energy chapter.

The parent already defines backward Euler for a more general signed CGK profile. Under fixed passive coefficients its storage change includes

$$-h(v^+)^T Cv^+ - \frac{1}{2}(v^+ - v)^T M(v^+ - v) - \frac{1}{2}(q^+ - q)^T K(q^+ - q) \quad (6)$$

in the unforced case. The last two terms are numerical dissipation from the selected update. Exact arithmetic preserves those terms exactly; it does not remove them. They cannot be reassigned to measured physical heat without a separate modeling decision. R13 therefore names an alternative finite profile.

2.2 PHYS-MIDPOINT-LINEAR-R1

For $h \geq 0$, write $\bar{q} = (q^+ + q)/2$ and $\bar{v} = (v^+ + v)/2$. Define

$$q^+ - q = h\bar{v}, \quad M(v^+ - v) = h(\bar{f} - K\bar{q} - C\bar{v}). \quad (7)$$

These undivided equations also define the $h = 0$ identity step. Eliminate q^+ to obtain the explicit finite solve

$$A_h = M + \frac{h}{2}C + \frac{h^2}{4}K, \quad (8)$$

$$b_h = (M - \frac{h}{2}C - \frac{h^2}{4}K)v - hKq + h\bar{f}, \quad (9)$$

$$v^+ = A_h^{-1}b_h, \quad q^+ = q + \frac{h}{2}(v^+ + v). \quad (10)$$

Proposition 2.1 (Existence and exact field closure). *For the stated symmetric sign conditions and $h \geq 0$, A_h is positive definite. Consequently the step is uniquely defined. If all supplied entries are rational, both output vectors have exact rational entries.*

Proof. For $w \neq 0$, $w^T A_h w = w^T M w + (h/2)w^T C w + (h^2/4)w^T K w > 0$. Thus A_h is invertible. A finite nonsingular linear solve over the rationals uses rational field operations and has rational output. No real root or transcendental decision is required for this selected rational profile. $\square$

Proposition 2.2 (Discrete power balance). *For the same fixed model,*

$$E(q^+, v^+) - E(q, v) = h\bar{v}^T \bar{f} - h\bar{v}^T C \bar{v}. \quad (11)$$

In particular, unforced undamped storage is preserved exactly, while the unforced damped step cannot increase it.

Proof. Symmetry gives $\Delta E = \bar{v}^T M(v^+ - v) + \bar{q}^T K(q^+ - q)$. Substitute (7); the stiffness terms cancel, leaving (11). Nonnegativity follows from $C \geq 0$. $\square$

The supplied work $h\bar{v}^T \bar{f}$ and dissipated energy $h\bar{v}^T C \bar{v}$ are the discrete profile's quantities. They are not automatically the exact work and heat integrals of the continuous trajectory. The theorem is a storage identity, not a proof of arbitrary-step trajectory accuracy or a guarantee about every other invariant.

2.3 Exact seed body and causal integration

The template arguments are

$$(q, v, M, C, K, h, \bar{f}) \mapsto (q^+, v^+, W_{\text{in}}, Q_d), \quad W_{\text{in}} = h\bar{v}^T \bar{f}, \quad Q_d = h\bar{v}^T C \bar{v}. \quad (12)$$

Use existing scalar arithmetic and matrix/vector construction/contraction, then `SOLVE_LINEAR` (opcode 87), to realize (10). The field operations, constants $1/2, 1/4$, and input lineage remain exact expressions. The output ledger is part of the template contract. No proof engine or compiled implementation is supplied here.

All operands read the same old snapshot. A completed event may select the interval's parameters according to the existing causal word policy. If that selection changes stiffness or other storage parameters, first account for the switch at the old state, then apply the fixed-parameter step. The force input and its actuator energy source stay explicit. Both output state roots commit together, and any rounded display remains an export.

This narrow profile makes no claim about time-varying mass, moving bases, nonlinear elasticity, contact, nonlinear reporter kinetics, orbital RK4 or arbitrary signed damping. Those applications retain their own definitions. Nonzero algorithmic residuals in another profile are reported as such; the word “exact” cannot turn them into a conservation proof.

2.4 Law residual as an output certificate

For the selected step, define the exact expression

$$r_E = E(q^+, v^+) - E(q, v) - W_{\text{in}} + Q_d. \quad (13)$$

The preceding algebra proves $r_E = 0$ on the template domain. A future compiler can use that identity as a rewrite or verification obligation without numerically comparing two nearly equal energies. If an implementation returns rounded state, its measured residual concerns that projection and cannot replace the expression-level proof. A residual of zero also does not establish that the physical parameters or observed device trajectory are correct.

3 Momentum, energy and admissible physical coupling

The exact seed needs a rule for composing physical models without creating energy at their interfaces. R13 adopts established momentum and thermodynamic balances, then derives a finite energy-port contract compatible with R12's mechanical and thermal states. These are restrictions on a selected physical model. They do not assign physical meaning to a bit pattern by themselves.

3.1 Momentum uses a realized frame and a declared body

For a closed material body in classical continuum mechanics, let $\rho > 0$ be mass density, v velocity, b body force per unit mass, and σ Cauchy stress. In an inertial Cartesian frame the local balance laws are

$$\dot{\rho} + \rho \nabla \cdot v = 0, \quad \rho \dot{v} = \nabla \cdot \sigma + \rho b, \quad \sigma = \sigma^T. \quad (14)$$

Dots here are material derivatives. Stress symmetry assumes the ordinary continuum without independent body couples or couple stresses. A moving control volume instead requires the corresponding mass and momentum fluxes. These balances and their energy/entropy counterparts are stated in [EN1: Banerjee, Basic Thermoelasticity, sections 1.1–1.3](#).

For a rigid body of fixed mass m , centre of mass r_C , body-to-inertial rotation $R_B \in SO(3)$ and constant body-frame inertia $I_B > 0$, the corresponding finite specialization is

$$m \ddot{r}_C = F_{\text{ext}}, \quad I_B \dot{\omega}_B + \omega_B \times (I_B \omega_B) = \tau_B, \quad (15)$$

$$\dot{R}_B = R_B [\omega_B]_{\times}, \quad [\omega_B]_{\times} a = \omega_B \times a. \quad (16)$$

The torque is about the centre of mass; body components and inertial components must not be mixed. The cross term follows from differentiating angular momentum in rotating axes. See [EN2: MIT Dynamics, lecture 28, Euler's equations](#). For this specialization direct contraction gives

$$\frac{d}{dt} \left(\frac{1}{2} m |\dot{r}_C|^2 + \frac{1}{2} \omega_B^T I_B \omega_B \right) = F_{\text{ext}} \cdot \dot{r}_C + \tau_B \cdot \omega_B, \quad (17)$$

because $\omega_B \cdot [\omega_B \times (I_B \omega_B)] = 0$. An inertia estimate, applied torque and orientation measurement are different inputs; a Madgwick attitude estimate alone supplies no torque law.

R12's frame annotations preserve coordinate identity. Their physical realization also requires origins, alignment and time convention. Transforming to accelerating or rotating coordinates changes the acceleration equation; renaming an inertial force law as a station-frame law is insufficient. Likewise a superposed rigid rotation must not generate elastic strain energy. For a finite-deformation hyperelastic model with deformation gradient $F = \nabla_X x$, $\det F > 0$, the selected objective free energy obeys

$$\Psi(QF, T) = \Psi(F, T), \quad Q \in SO(3), \quad (QF)^T(QF) = F^T F. \quad (18)$$

Material directions, if present, remain explicit reference data; objectivity does not imply isotropy. This condition and the two-point transformation of F are discussed in [EN3: Bouchbinder, Non-Equilibrium Continuum Physics, section 4](#). The existing $\epsilon = \text{sym } \nabla u$ model remains a small-gradient approximation; equation (18) does not extend it to arbitrary finite rotations. A finite-rotation specialization must supply its own objective constitutive equation and compatible finite state.

3.2 A finite port contract with an explicit power proof

Use the finite state x , differentiable storage $H(x)$, and matrices $\mathcal{J}(x) = -\mathcal{J}(x)^T$, $\mathcal{R}(x) = \mathcal{R}(x)^T \geq 0$ and $\mathcal{G}(x)$. Define

$$\dot{x} = (\mathcal{J} - \mathcal{R}) \nabla H + \mathcal{G} u, \quad y = \mathcal{G}^T \nabla H. \quad (19)$$

This is the standard finite port-Hamiltonian form; $u^T y$ is power supplied at the declared boundary. See EN4: [van der Schaft, Port-Hamiltonian nonlinear systems, equations 5–6](#). The symbol $\mathcal{J}$ is unrelated to the JK input word J . Each contraction must be dimensionally compatible. If components of x have different units, their derivative and matrix entries have corresponding component units; they are stored as typed tuples and explicit contractions, not forced into a homogeneous vector type.

Proposition 3.1 (Finite power balance and composition). *For a differentiable solution of (19),*

$$\dot{H} = u^T y - \mathcal{D}, \quad \mathcal{D} = (\nabla H)^T \mathcal{R} \nabla H \geq 0. \quad (20)$$

For two such systems whose ports admit a dimensionally compatible map L , connect $u_1 = Ly_2$, $u_2 = -L^T y_1$. Their internal power cancels. Their combined storage is $H_1 + H_2$ and its rate is minus their dissipation plus any remaining external port power.

Proof. Apply the chain rule and use $z^T \mathcal{J} z = 0$ for a skew matrix. The two connected powers sum to $y_2^T L^T y_1 - y_1^T L y_2 = 0$. The entries of L carry any required unit and frame conversion; a force is not silently identified with a velocity. $\square$

If H has a lower bound, integration gives the usual passivity inequality $H(t) - H(0) \leq \int_0^t u^T y \, ds$. This limits extraction relative to storage and supply; it does not by itself prove asymptotic stability, accurate tracking, or a sufficient actuator. Algebraic port connections must also yield a well-defined finite evolution.

Physical port pairs include force/velocity, torque/angular velocity and voltage/current. In a thermal port the energy pairing is absolute temperature times entropy flow, $T\dot{S}_{\text{transfer}} = \dot{Q}$ at that port. Pairing temperature directly with heat power and calling the product power would have the wrong units. Entropy production inside the body is distinct from entropy transported through a port.

3.3 The existing modal mechanics fits this contract

R12 already declares a finite displacement expansion $u_m(X, t) = \sum_i q_i(t) \phi_i(X)$, mass and stiffness matrices from its weak form, and support conditions. For a fixed basis, fixed material parameters, isothermal or explicitly uncoupled mechanics, take

$$M\ddot{q} + D\dot{q} + Kq = Bu, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad D = D^T \geq 0. \quad (21)$$

Let $p = M\dot{q}$ and

$$H_m(q, p) = \frac{1}{2} p^T M^{-1} p + \frac{1}{2} q^T K q, \quad \mathcal{J} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathcal{R} = \begin{bmatrix} 0 & 0 \\ 0 & D \end{bmatrix}, \quad \mathcal{G} = \begin{bmatrix} 0 \\ B \end{bmatrix}. \quad (22)$$

Then (19) gives exactly (21), with

$$y = B^T \dot{q}, \quad \dot{H}_m = u^T B^T \dot{q} - \dot{q}^T D \dot{q}. \quad (23)$$

The PSD stiffness admits free modes; asymptotic displacement stability does not follow when rigid or undamped modes remain. If M depends on configuration, the derivative of its kinetic energy is part of the dynamics; merely replacing constant M by $M(q)$ in (21) generally misses terms.

Finite modal reduction preserves this displayed energy identity when its coefficient assumptions hold. It does not prove the basis resolves every physical mode. Spatial integrals still require the charts, measures and integrability obligations already specified by R12. A continuum PDE printed above is an origin for the model; it is not an implemented infinite-dimensional XOP operator.

An actual ASA/NA–JK specialization provides an explicit decode from its completed old state to a command, then a transduction law to the physical effort $u(t)$. A voltage command cannot be substituted as a force without that law. The resulting work is

$$W_n = \int_{t_n}^{t_{n+1}} u(t)^T y(t) dt. \quad (24)$$

The command may be held over the interval; the velocity generally is not. A bit selecting a positive effort can supply energy through the actuator's power source. Neither bit packing nor a delayed update makes that port passive.

3.4 Convert dissipated work to heat exactly once

For the uncoupled mechanical storage above let $P_d = \dot{q}^T D \dot{q} \geq 0$. Suppose all this dissipated mechanical power thermalizes in the modeled assembly, and fixed or state-declared fractions $w_i \geq 0$, $\sum_i w_i = 1$, distribute it among thermal nodes. Write

$$\dot{H}_m = P_{\text{mech,in}} - P_d, \quad (25)$$

$$C_i \dot{T}_i = P_{i,\text{other}} + w_i P_d + \sum_{j \neq i} G_{ij}(T_j - T_i) + G_{i0}(T_a - T_i), \quad (26)$$

where $C_i > 0$, $G_{ij} = G_{ji} \geq 0$ and $G_{i0} \geq 0$. The stores $U_i = C_i(T_i - T_{i,\text{ref}})$ are an uncoupled constant-capacity model. Their sum with H_m obeys the directly derived identity

$$\frac{d}{dt}(H_m + \sum_i U_i) = P_{\text{mech,in}} + \sum_i P_{i,\text{other}} + \sum_i G_{i0}(T_a - T_i). \quad (27)$$

Internal conduction cancels in pairs, as does mechanical-to-thermal power. Thus P_d is not added again as an external input. If only a fraction thermalizes locally, the remaining energy needs its destination, such as outgoing sound or a separate store. A resistor's VI and an optical absorber's thermalized energy are handled with the same nonduplicating boundary ledger. The distinction between work conversion and entropy creation is treated in [EN5: MIT, Work-to-heat Transduction](#).

3.5 Entropy supplies a second, independent condition

Conserving total energy does not make every proposed heat exchange physical. Restrict the node model to local equilibrium with $T_i > 0$, and define

$$S_i - S_{i,\text{ref}} = C_i \ln(T_i/T_{i,\text{ref}}), \quad T_{i,\text{ref}} > 0, \quad \dot{S}_i = C_i \dot{T}_i/T_i. \quad (28)$$

For one reciprocal conductive link the derived total entropy rate is

$$\dot{S}_{ij} = \frac{G_{ij}(T_j - T_i)}{T_i} + \frac{G_{ij}(T_i - T_j)}{T_j} = \frac{G_{ij}(T_i - T_j)^2}{T_i T_j} \geq 0. \quad (29)$$

A link to an ideal bath at $T_a > 0$ has the same expression after including the bath's entropy change. Local thermalization of $w_i P_d$ contributes $w_i P_d/T_i \geq 0$. Arbitrary external heat, light, matter and work ports must carry their own energy and entropy accounting; an incident optical field's entropy flux is not assigned the thermal formula P_{optical}/T_i merely because its energy is later absorbed.

In local continuum heat conduction, with heat flux $q_h = -\kappa \nabla T$ and positive-semidefinite symmetric conductivity κ , the analogous production density is

$$\xi_{\text{cond}} = q_h \cdot \nabla(1/T) = \frac{(\nabla T)^T \kappa \nabla T}{T^2} \geq 0. \quad (30)$$

Fourier's constitutive law is a local-equilibrium model, not a universal description of ballistic transport or ultrafast material response. Entropy flux and production are distinguished in [EN6: MIT, Entropy Production and Nonlinearity](#) and [EN7: MIT Advanced Thermodynamics, lecture 21, heat conduction](#).

3.6 Thermoelastic expansion requires reciprocal heat coupling

R12's thermal strain is useful, but a coupled model cannot independently choose an elastic energy, expansion force and unrelated heat capacity. The following small-gradient, local-equilibrium specialization derives them from one Helmholtz free-energy density $\psi(\epsilon, T)$ per reference volume. All densities below use that same small-strain reference convention. Define

$$s = -\psi_T, \quad e = \psi + Ts = \psi - T\psi_T, \quad \sigma = \psi_\epsilon + \sigma_d, \quad c_\epsilon = -T\psi_{TT} > 0. \quad (31)$$

Here s is entropy density, e internal energy density, σ_d dissipative stress, and c_ϵ volumetric heat capacity at fixed strain. These thermodynamic conjugacies follow the free-energy formulation in [EN1, sections 1.3.3–1.3.4](#); the following compatibility calculation is the R13 specialization.

Combining the local balances $\dot{e} = \sigma : \dot{\epsilon} - \nabla \cdot q_h + r$ and $\dot{s} + \nabla \cdot (q_h/T) - r/T = \xi \geq 0$ gives

$$T\xi = \sigma : \dot{\epsilon} - \dot{\psi} - s\dot{T} - q_h \cdot \nabla T/T, \quad (32)$$

$$= \sigma_d : \dot{\epsilon} - q_h \cdot \nabla T/T \geq 0, \quad (33)$$

$$c_\epsilon \dot{T} = T\psi_{T\epsilon} : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (34)$$

The last equation follows by differentiating $e = \psi - T\psi_T$ and canceling $\psi_\epsilon : \dot{\epsilon}$. Here r is heat supply after any declared local conversion, rather than a second copy of mechanical dissipation. With $\sigma_d = \mathbb{V} : \dot{\epsilon}$, symmetric $\mathbb{V} \geq 0$ on symmetric strains, and Fourier conductivity as above, both displayed dissipative contributions are nonnegative. The reversible term $T\psi_{T\epsilon} : \dot{\epsilon}$ need not be positive: it exchanges thermal and mechanical storage without being dissipation.

A concrete minimal profile uses constant elastic tensor $\mathbb{C}$ positive on admissible strains, constant symmetric thermal-stress tensor β , constant $c_v > 0$ and $T_0 > 0$:

$$\psi(\epsilon, T) = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon - (T - T_0)\beta : \epsilon + c_v [T - T_0 - T \ln(T/T_0)], \quad (35)$$

$$\sigma = \mathbb{C} : \epsilon - (T - T_0)\beta + \sigma_d, \quad (36)$$

$$s = \beta : \epsilon + c_v \ln(T/T_0), \quad (37)$$

$$e = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon + T_0\beta : \epsilon + c_v(T - T_0), \quad c_\epsilon = c_v, \quad (38)$$

$$c_v \dot{T} = -T\beta : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (39)$$

Choosing $\beta = \mathbb{C} : (\alpha_T I)$ reproduces R12's constant-coefficient thermal-expansion stress. The linear term in e is required by the selected entropy convention and coupling; replacing e by an unrelated "elastic plus $c_v T$ " formula breaks the same first-law calculation. The profile applies on a declared strain and positive-temperature range; it is not a fitted constitutive law for an unspecified specimen.

Additional reporter, damage or chemical state z changes this closure to $\psi(\epsilon, T, z)$ and adds $-\psi_z \cdot \dot{z}$ to $T\xi$. An interval-preserving reporter update from R12 proves its range, not this new thermodynamic inequality. Either specify the driving energy/rates that meet it, or keep the reporter as an empirical readout state with its energy exchange separately declared. No new chemical model is assumed here.

3.7 Switching and exact steps keep their own energy obligations

If a physical mode a changes the storage function, continuous parameter motion has the additional power term $H_a \dot{a}$ in the chain rule. At an idealized discrete switch, use one consistent reference for the stores and record

$$\Delta H_{\text{switch}} = H_{a^+}(x^+) - H_{a^-}(x^-) = W_{\text{switch,in}} - E_{\text{switch,out}}. \quad (40)$$

An ideal routing switch can have zero storage jump. A change of stiffness at fixed q, p and fixed M instead contributes $\frac{1}{2}q^T(K^+ - K^-)q$; a state reset or impulsive motion has its own mechanical

balance. Labeling the selector as ASA, NA, JK or a thermal bit does not remove these terms. Passive-mode energy inequalities cannot simply be concatenated across unaccounted jumps.

Exact evaluation preserves the expression of a selected time-step map. It does not automatically preserve (20) between its endpoints. Section 2 supplies a finite midpoint specialization with its own derived step energy identity for fixed linear mechanical coefficients. The continuous coupled thermoelastic equations here retain their separate finite-reduction, existence and time-discretization obligations.

3.8 What is added to the kernel contract

No opcode is needed for these laws. R13 binds energy and entropy functions, their derivatives, domain conditions, port pairings and selected finite evolution as versioned templates using existing exact arithmetic, LN, finite integrals and finite ODE declarations. States of unlike units use tuples; matrix contractions expand componentwise. A valid specialization can now present its power identity, heat-transfer cancellation and nonnegative entropy production as explicit mathematical obligations. A false sign, unpowered actuator, duplicated heat term or incompatible thermal expansion can be identified before a kernel is run. This is the useful added foundation. Specimen parameters, model adequacy, runtime evaluation and physical performance remain separate evidence.

4 Electromagnetic closure of the optical interface

R12 defines $U_{\text{det},\lambda} = H_{z,\lambda}[A t_\lambda e^{i\phi_{\text{lens},\lambda}} U_{\text{in},\lambda}]$. This chapter supplies a physical route to its transfer, loss and material-state arguments. Maxwell's equations are the adopted classical field laws; a material law, geometry and boundary data select a particular device. The added templates fit the existing exact expression and finite-state interfaces.

4.1 Fields, charge and the data that close their evolution

Use SI quantities in one declared spatial frame and time coordinate. For a stationary macroscopic medium, with free charge ρ_f and free current $\mathbf{J}_f$, adopt

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}, & \nabla \times \mathbf{H} &= \mathbf{J}_f + \partial_t \mathbf{D}. \end{aligned} \quad (41)$$

The dimensions are $[\mathbf{E}] = \text{V/m}$, $[\mathbf{H}] = \text{A/m}$, $[\mathbf{D}] = \text{C/m}^2$, $[\mathbf{B}] = \text{T}$, $[\rho_f] = \text{C/m}^3$ and $[\mathbf{J}_f] = \text{A/m}^2$. The divergence of the last equation gives

$$\partial_t \rho_f + \nabla \cdot \mathbf{J}_f = 0. \quad (42)$$

Thus supplied charge and current histories cannot be chosen independently. The electric divergence constraint propagates when it holds initially and the supplied charge obeys continuity; the magnetic constraint propagates because the divergence of a curl vanishes. These statements require the corresponding classical derivatives or an explicitly declared weak solution. The laws and charge-conservation argument are documented in the [Caltech Feynman Lectures, II.18 \[EM1\]](#).

For a stationary interface, let $\mathbf{n}$ point from medium 1 to medium 2. With surface free charge ρ_s , surface free current $\mathbf{K}_s$, and no magnetic surface sources, the interface conditions are

$$\begin{aligned} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= \rho_s, & \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) &= 0, \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) &= 0, & \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) &= \mathbf{K}_s. \end{aligned} \quad (43)$$

They follow by shrinking Maxwell's integral pillboxes and loops across the interface. A moving boundary needs the corresponding moving-medium model; these stationary formulas cannot silently serve as that model. [MIT Electromagnetics and Applications, Section 2.6 \[EM2\]](#).

A device profile binds its material regions, interface geometry, constitutive relations, sources, compatible initial fields and initial material memory. It selects the needed independent boundary or incoming-wave data, including an outgoing-wave condition for an open problem. It does not prescribe every field component independently on every boundary. Existence and uniqueness belong to this completed problem; the four field equations alone do not resolve a resonant, contradictory or underspecified setup.

4.2 Power balance before optical loss is called heat

The vector identity for $\nabla \cdot (\mathbf{E} \times \mathbf{H})$ and equation (41) give the constitutive-independent identity

$$\nabla \cdot \mathbf{S} + \mathbf{E} \cdot \partial_t \mathbf{D} + \mathbf{H} \cdot \partial_t \mathbf{B} = -\mathbf{E} \cdot \mathbf{J}_f, \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (44)$$

For instantaneous, time-independent real symmetric positive-definite tensors ϵ, μ , with $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$, set

$$u_{\text{em}} = \frac{1}{2} \mathbf{E} \cdot \epsilon \mathbf{E} + \frac{1}{2} \mathbf{H} \cdot \mu \mathbf{H},$$

$$\partial_t u_{\text{em}} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_f. \quad (45)$$

The quadratic energy follows by differentiating each form; time independence and symmetry are used at that exact step. This is Poynting's energy balance, with fields doing work on matter through $\mathbf{E} \cdot \mathbf{J}_f$. [Caltech Feynman Lectures, II.27 \[EM3\]](#).

Writing $\mathbf{J}_f = \sigma \mathbf{E} + \mathbf{J}_{\text{src}}$, where the symmetric part of σ is positive semidefinite, identifies a nonnegative Ohmic sink $q_{\text{Ohm}} = \mathbf{E} \cdot \sigma \mathbf{E}$. The impressed source term $-\mathbf{E} \cdot \mathbf{J}_{\text{src}}$ may supply field energy. In contrast, arbitrary $\mathbf{E} \cdot \mathbf{J}_f$ need not be positive or entirely heat: work can enter mechanical or other material stores. A dispersive dielectric also cannot use $\frac{1}{2} \mathbf{E} \cdot \mathbf{D}$ as its complete instantaneous energy without a justified material-state model.

4.3 A finite dispersive material law with an energy proof

An optional, explicit local linear closure uses finitely many polarization oscillators. In a fixed material region let $\epsilon_b, \mu > 0$ be scalar, time-independent background coefficients and set

$$\mathbf{D} = \epsilon_b \mathbf{E} + \sum_{j=1}^m \mathbf{P}_j, \quad \mathbf{B} = \mu \mathbf{H},$$

$$\dot{\mathbf{P}}_j = \mathbf{V}_j, \quad \dot{\mathbf{V}}_j = f_j \mathbf{E} - \gamma_j \mathbf{V}_j - \omega_j^2 \mathbf{P}_j, \quad (46)$$

where m is finite, $f_j > 0$, $\gamma_j \geq 0$ and $\omega_j > 0$. Here $[f_j] = [\epsilon_b]/s^2$, $[\gamma_j] = s^{-1}$ and $[\omega_j] = s^{-1}$. Each oscillator has supplied $\mathbf{P}_j(t_0), \mathbf{V}_j(t_0)$. This is a selected Lorentz material model, not a universal constitutive law. Its optical origin is described in [MIT Fundamentals of Photonics, Section 2.1.4 \[EM4\]](#).

Proposition 4.1 (Explicit storage and dissipation). *For the held coefficients in equation (46), define*

$$u_j = \frac{|\mathbf{V}_j|^2 + \omega_j^2 |\mathbf{P}_j|^2}{2f_j},$$

$$u_* = \frac{1}{2} \epsilon_b |\mathbf{E}|^2 + \frac{1}{2} \mu |\mathbf{H}|^2 + \sum_j u_j,$$

$$q_{\text{diss}} = \mathbf{E} \cdot \sigma \mathbf{E} + \sum_j \frac{\gamma_j}{f_j} |\mathbf{V}_j|^2 \geq 0. \quad (47)$$

Then $\partial_t u_* + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_{\text{src}} - q_{\text{diss}}$.

Proof. Multiplying the oscillator equation by $\mathbf{V}_j/f_j$ gives $\dot{u}_j = \mathbf{E} \cdot \mathbf{V}_j - \gamma_j |\mathbf{V}_j|^2/f_j$. In equation (44), the polarization part of $\mathbf{E} \cdot \dot{\mathbf{D}}$ is $\sum_j \mathbf{E} \cdot \mathbf{V}_j$. Adding the oscillator balances cancels these exchanges exactly. Positivity follows from the coefficient and conductivity hypotheses. $\square$

This supplies a consistent source for R12's thermal ledger. Assign $P_{\text{heat,opt}} = \int_{\Omega} q_{\text{diss}} dV$ only when these dissipative channels thermalize in the modeled thermal store on the declared interval. If excitation, emission or another store participates, partition the sink and retain that store's evolution. Counting the same absorbed power independently as full heat and full chemical excitation would break the balance.

With the phasor convention $\mathbf{E}(t) = \text{Re}(\widehat{\mathbf{E}}e^{-i\omega t})$, $\omega > 0$, the scalar isotropic, steady harmonic closure is

$$\epsilon_{\text{eff}}(\omega) = \epsilon_b + \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} + \frac{i\sigma}{\omega},$$

$$\langle q_{\text{diss}} \rangle = \frac{\omega}{2} \text{Im}(\epsilon_{\text{eff}}) |\widehat{\mathbf{E}}|^2. \quad (48)$$

All denominators must be nonzero, and the selected steady response must exist; an undamped resonance does not satisfy these assumptions. Conductivity is included once, inside ϵ_{eff} in this frequency convention. The time-domain state in equation (46) remains the definition when a steady response is inappropriate.

Temperature, hysteresis and reporter state may parameterize a family of these material profiles. Holding them fixed during an optical subproblem is an explicit time-scale approximation. If coefficients change during propagation, their derivatives change stored energy: for example, $\partial_t(\epsilon_b |\mathbf{E}|^2/2)$ includes $\dot{\epsilon}_b |\mathbf{E}|^2/2$. The corresponding modulation work and material exchange must be included. Replacing a coefficient at each tick without that accounting is not the proved coupled energy law.

4.4 How Maxwell closes the existing propagation operator

In a homogeneous, source-free, lossless region with constant scalar $\epsilon, \mu > 0$, the divergence constraint gives $\nabla \cdot \mathbf{E} = 0$. Taking a curl and using $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ yields

$$\nabla^2 \mathbf{E} - \mu\epsilon \partial_t^2 \mathbf{E} = 0,$$

$$(\nabla^2 + k^2) \widehat{\mathbf{E}} = 0, \quad k = \omega \sqrt{\mu\epsilon}. \quad (49)$$

For a linear dispersive homogeneous medium, use the corresponding frequency equation with its declared complex constitutive coefficients and radiation branch. A scalar component can stand for the optical field only within an appropriate polarization/geometry reduction. Spatially varying tensors, strong polarization conversion or near-field boundary interactions do not automatically reduce to the same scalar equation. [MIT Optics, Lecture 9 \[EM5\]](#).

For the scalar, forward, homogeneous lossless case write $U = e^{ikz}\psi$. Substitution gives

$$\partial_z^2 \psi + 2ik\partial_z \psi + \nabla_{\perp}^2 \psi = 0. \quad (50)$$

The paraxial approximation drops $\partial_z^2 \psi$ when it is negligible relative to the retained terms on the admitted field class. For $z > 0$, one resulting Fresnel transfer, with $\lambda_m = 2\pi/k$, is

$$(H_z U_0)(x, y) = \frac{e^{ikz}}{i\lambda_m z} \iint U_0(x', y') \exp\left(\frac{ik[(x-x')^2 + (y-y')^2]}{2z}\right) dx' dy'. \quad (51)$$

This states the normalization missing from a mere angular-shape transform. For R12's finite aperture, the integral is over its bounded support; its exact real-pair lift needs the section integrability conditions already specified there. A full-plane operator has its separately declared function-space domain. Fraunhofer additionally neglects the aperture-plane quadratic phase on the stated field/support scale. These approximations need application error allowances, not an assertion that exact encoding makes them exact Maxwell solutions. [MIT Optics, Lecture 16 \[EM6\]](#).

The same boundary-value problem fixes t_λ and reflected/outgoing channels from material parameters and geometry. A thin local multiplier is an adopted reduction; a general scattering device may require a matrix or integral transfer. The aperture A remains a geometric predicate. It does not specify the polarization currents or electromagnetic boundary response of the physical opaque material by itself.

4.5 Passivity at the same physical interface

For a fixed volume, let $\mathcal{U} = \int_{\Omega} u_* dV$. Separate its boundary flux into admitted incoming and outgoing power. With no impressed source, integration of the derived energy identity gives

$$E_{\text{out}}[t_0, t_1] \leq E_{\text{in}}[t_0, t_1] + \mathcal{U}(t_0) - \mathcal{U}(t_1). \quad (52)$$

Stored energy can temporarily power an output. In steady harmonic operation, the average storage change vanishes; for independent positive-flux modal coordinates this gives precisely R12's condition

$$S^\dagger W_{\text{out}} S \leq W_{\text{in}}. \quad (53)$$

Both metrics refer to the same physical normalization and selected ports. Unobserved reflected, radiated or coupled-away power is not automatically absorption. Only with all outgoing channels accounted for, no supplied gain and no net storage accumulation does $\mathcal{R} + \mathcal{T} + \mathcal{A} = 1$ identify the remaining fraction as material absorption. This derivation supplies the physical premise for the R12 scaled passive pinion construction; it does not change its algebra or demonstrate a manufactured device.

One optional mechanical coupling is already useful without a general electromagnetic stress model. For a stationary planar target in vacuum, normal incidence, forward transmission and specular backward reflection, the steady force along the incident beam is

$$F_{\parallel} = \frac{P_{\text{in}}}{c}(1 + \mathcal{R} - \mathcal{T}) = \frac{P_{\text{in}}}{c}(\mathcal{A} + 2\mathcal{R}). \quad (54)$$

It follows by subtracting outgoing from incoming momentum flux. It can enter the declared mechanical load. Oblique/diffuse scattering, moving media, directional re-emission or additional forces require their corresponding momentum balance; the scalar formula does not cover them. [Caltech Feynman Lectures, I.34.9 \[EM7\]](#).

4.6 Finite exact seed representation

Complex coefficients, amplitudes and matrices use R12's ordered real-pair templates. For a declared finite modal problem $Az = b$, with $A = X + iY$, $z = p + iq$ and $b = r + is$, the literal real system is

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} r \\ s \end{pmatrix}. \quad (55)$$

An inverse requires a proved nonzero determinant; an unresolved singularity or branch remains an explicit non-value outcome. The coefficients bind the chosen basis, geometry, boundary projection, material law and frequency. They may be retained exact expressions or supplied calibrated literals with uncertainty. For instance, select a bounded fixed cavity with a perfect electric conductor

boundary and real admissible basis functions ψ_j whose tangential traces vanish. The expansion $\widehat{\mathbf{E}}_N = \sum_{j=1}^N z_j \psi_j$ and weak projection of the frequency curl–curl equation give

$$\begin{aligned} A_{ab} &= \int_{\Omega} [(\nabla \times \psi_a) \cdot \mu^{-1}(\nabla \times \psi_b) - \omega^2 \psi_a \cdot \epsilon_{\text{eff}} \psi_b] dV, \\ b_a &= i\omega \int_{\Omega} \psi_a \cdot \widehat{\mathbf{J}}_{\text{src}} dV. \end{aligned} \quad (56)$$

The boundary term vanishes under this particular trace condition. The finite space, source projection, divergence constraints and coefficient integral domains must be specified. Other boundaries require their own terms. These volume-integral definitions fit existing exact operators only when their actual integrands and bounded coordinate domains meet that operator contract; the weak notation alone is not an executable integral encoding.

The finite polarization coordinates use the first-order equations in (46). Together with a declared finite field reduction, they form an ordinary finite state evolution only after building algebraic constraints into the basis or explicitly eliminating them to obtain a nonsingular finite evolution. The fixed representation, selected elimination branch and regular domain remain explicit. XOP can denote the resulting ODE under its existence/domain contract or an explicitly selected finite update. Remaining algebraic constraints need their own defined constrained formulation. It does not acquire an arbitrary continuum PDE solver. Basis truncation, omitted modes and an approximate constitutive fit remain model errors even when every retained operation and packed bit is exact. The physical profile therefore exports the field transfer, dissipative/other energy channels and optional force from one common model, rather than fitting unrelated outputs that happen to share a seed identifier.

5 Measurement, constants and output tolerances

The existing XOP dimension, frame, sample and dependency types already carry the information needed here. This chapter adds a physical interpretation and conditional output certificates, without adding a measurement oracle or a new packet format. An exactly stored indication, an exact nominal model and the physical quantity being inferred remain distinct. The purpose is operational: preserve what is known, keep shared uncertainties coupled, and determine what an output can claim within a selected application tolerance.

5.1 Exact defining constants and measured coefficients

Only four SI defining constants are needed by the present optical, electrical and thermal profiles. Their numerical values are fixed by the SI; unit realizations and measurements made with them still have uncertainty. The following dimension vectors use the inherited order $(L, M, T, I, \Theta, N, J_{\text{lum}})$. [\[ME1: BIPM defining constants\]](#)

Constant	Exact SI quantity	XOP dimension vector
c	$299\,792\,458 \text{ m s}^{-1}$	$(1, 0, -1, 0, 0, 0, 0)$
h	$6.626\,070\,15 \times 10^{-34} \text{ J s}$	$(2, 1, -1, 0, 0, 0, 0)$
e	$1.602\,176\,634 \times 10^{-19} \text{ C}$	$(0, 0, 1, 1, 0, 0, 0)$
k_B	$1.380\,649 \times 10^{-23} \text{ J K}^{-1}$	$(2, 1, -2, 0, -1, 0, 0)$

The decimal coefficients are finite rational literals, reduced canonically by the R11 grammar. For example, the numerical coefficient of h is $662607015/10^{42}$ before reduction. A binary floating-point approximation is unnecessary. The elementary charge e is positive; the charge of an electron is $-e$. The namespace distinguishes it from the Boolean accepted-event bit.

The gravitational constant G and fine-structure constant α_{fs} are measured quantities, with dimensions $L^3 M^{-1} T^{-2}$ and 1, respectively. A specialization that needs them binds a dated value, uncertainty and provenance; this core does not hard-code a new estimate. [ME2: NIST CODATA table] In present SI, vacuum permeability and permittivity are also not independent exact decimal constants. For positive α_{fs} , retain their common dependency:

$$\mu_0 = \frac{2\alpha_{\text{fs}} h}{e^2 c}, \quad \epsilon_0 = \frac{e^2}{2\alpha_{\text{fs}} h c}, \quad \mu_0 \epsilon_0 = \frac{1}{c^2}. \quad (57)$$

Here $[\mu_0] = MLT^{-2}I^{-2}$ and $[\epsilon_0] = M^{-1}L^{-3}T^4I^2$. Multiplication cancels the shared α_{fs} exactly. Replacing the two occurrences by unrelated uncertain inputs would lose that constraint. The former exact assignment $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ is not the present SI definition. [ME3: SI Brochure, 9th edition, 2026, section 2.3.2]

5.2 An observation is a modeled acquisition

Let $x(t)$ be the selected physical state, θ its sensor/calibration parameters, and m the transduction law into an analogue indication. An averaging acquisition over $[t_-, t_+]$ has the model

$$M[x, \theta, w] = \int_{t_-}^{t_+} w(t) m(x(t), \theta, t) dt, \quad w(t) \geq 0, \quad \int_{t_-}^{t_+} w(t) dt = 1, \quad (58)$$

$$y = Q_{\Delta}(M[x, \theta, w] + d_M + \eta). \quad (59)$$

The weighting w has dimension T^{-1} ; y, M , the model discrepancy d_M and analogue noise η have the same indication dimension. For a counter or accumulated dose, use its declared accumulation law instead of a normalized average. An instantaneous sample is a separately defined profile, not an implicit substitution of the interval endpoint for the integral. Each integral requires finite bounds and its stated integrability conditions. The profile supplies finite function templates or explicit external bindings for the trajectory and weighting; it cannot encode an arbitrary unknown continuous signal as a fully known finite literal.

Q_{Δ} includes its step, offset, ties rule, range and saturation behavior. For an unsaturated uniform nearest quantizer with step $\Delta > 0$, put $v = M + d_M + \eta$. Then

$$y = M + d_M + \eta + q, \quad q = Q_{\Delta}(v) - v, \quad |q| \leq \Delta/2. \quad (60)$$

This is a deterministic bound, not an assumption that q is uniform, independent or newly random at every sample. Saturated codes require the actual preimage of the saturation law; the half-step bound no longer follows. The stored integer and its rational unit conversion can be exact while the corresponding physical indication occupies a range.

The measurement model also declares its valid specimen, operating conditions and reference conditions. A discrepancy bound is supplied evidence or a stated hypothesis, never a number generated merely by writing d_M . Unrecognized effects are not covered automatically. This makes explicit the model scope emphasized by JCGM GUM-6, sections 6–7. [ME4: Developing and using measurement models]

5.3 Calibration and joint uncertainty

Let p collect the finite parameters of the chosen profile: initial state, calibration, clock and frame realization, input history parameters and relevant residuals. A joint admissible set $\mathcal{U}$ retains their actual shared dependencies. With observation map F built from (59), the data-compatible set is

$$\mathcal{U}_y = \{p \in \mathcal{U} : F(p) = y\}, \quad \mathcal{Z}_y = \{g(p) : p \in \mathcal{U}_y\}. \quad (61)$$

For multiple records, impose all their observation equations on the same shared calibration parameters. A repeated common offset does not become a new independent offset. If the set is proved empty, the declared model, admissible bounds and observations are inconsistent; empty-set containment must never be used to certify an output. If solving it is unresolved, retain the unresolved status. Neither conclusion identifies which physical assumption failed.

A simple calibration model $y^{\text{cal}} = A\theta + \varepsilon$ illustrates what equations alone can establish. With known design matrix $A \in \mathbb{R}^{m \times r}$, for consistent noiseless data the parameters are unique exactly when

$$\text{rank } A = r. \quad (62)$$

For a chosen positive-definite covariance Σ and full column rank, the generalized least-squares expression is

$$\hat{\theta} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} y^{\text{cal}}. \quad (63)$$

The inverse exists since $z^T A^T \Sigma^{-1} A z = (Az)^T \Sigma^{-1} (Az) > 0$ for $z \neq 0$. This proves the algebraic expression's domain, not a calibration accuracy. Uncertain reference settings make A uncertain and must enter the joint model. For a differentiable nonlinear calibration, a full-column-rank Jacobian permits a local regular inverse by selecting independent observation components. It does not give global uniqueness. Rank deficiency alone does not decide nonlinear identifiability.

An alternative statistical profile supplies a joint distribution or covariance with its basis. For a differentiable output map and its linearization $J = Dg(\hat{p})$,

$$\Sigma_z \simeq J \Sigma_p J^T, \quad u^2(z_i) \simeq \sum_{j,k} J_{ij} (\Sigma_p)_{jk} J_{ik}. \quad (64)$$

Cross terms retain correlations. This is exact for an affine map with the stated covariance; in general it is a first-order approximation and a standard uncertainty is not a worst-case bound. [ME5: JCGM 100, sections 5.1–5.2] Significant nonlinearity can also affect the output estimate itself; exact evaluation of $g(\hat{p})$ does not make it the distributional mean. [ME6: JCGM 100, Amendment 1:2026]

5.4 Acquisition time and realized coordinates

R12's WI record, immutable registry and session consumption rules remain unchanged. The calibration identifier resolves complete calibration content; its presence is not proof of freshness or traceability. Add the chosen measurement model, acquisition weighting, quantizer and joint-uncertainty definition to that existing dependency closure.

For recorded source times $\tau_{\pm}$, a declared affine clock model is

$$t_{\pm} = a\tau_{\pm} + b + \delta t_{\pm}, \quad a > 0. \quad (65)$$

The same a, b appear at both endpoints and in every record covered by that calibration. The physical acquisition interval, conversion validity interval and availability time remain distinct. Subtracting endpoints cancels the shared offset b ; treating it as two independent offsets would miss this exact dependency. The uncertainty of a and endpoint residuals remains.

The analogous physical frame map is

$$x_B(t) = R_{BA}(t)x_A(t) + o_{BA}(t), \quad R_{BA}^T R_{BA} = I, \quad \det R_{BA} = 1. \quad (66)$$

The XOP frame identity specifies the convention; R_{BA} and o_{BA} specify its realization. A shared uncertain mounting is retained once in p , and an interval enclosure of its entries must retain, or conservatively enclose, the rotation constraints. A nominal matrix is not made physically aligned merely by encoding it exactly. Coordinate values from different acquisition times require the selected dynamical/time-transfer model before they can represent simultaneous geometry.

5.5 A bounded output certificate

The core needs a reusable rule for deciding a functional tolerance without inventing a universal numerical tolerance. Let K be a nonempty compact convex set in the finite parameter coordinates; require $\hat{p} \in K$ and a nonempty $\mathcal{U}_y \subseteq K$. The output g_i is continuously differentiable on an open neighborhood of K . Suppose the following finite bounds have been established for all $p \in K$:

$$|p_j - \hat{p}_j| \leq r_j, \quad \left| \frac{\partial g_i}{\partial p_j}(p) \right| \leq L_{ij}. \quad (67)$$

L_{ij} has output-unit per parameter-unit dimensions. No covariance or independence assumption is used. By integrating the derivative along the segment from $\hat{p}$ to p ,

$$g_i(p) - g_i(\hat{p}) = \int_0^1 \sum_j \frac{\partial g_i}{\partial p_j}(\hat{p} + s(p - \hat{p}))(p_j - \hat{p}_j) ds, \quad (68)$$

$$|g_i(p) - g_i(\hat{p})| \leq \sum_j L_{ij} r_j. \quad (69)$$

The compact domain stays clear of undefined divisions, branch changes or other points where the derivative hypotheses fail. A box can conservatively enclose a correlated set; using the original joint constraints can tighten the result. Piecewise or discrete outputs use direct image/branch enclosure instead of differentiating through a switch.

To apply the certificate to a physical result, assume its actual parameter vector lies in $\mathcal{U}_y$. Nonemptiness alone does not establish that physical inclusion. Suppose the intended physical output is $z_i^{\text{phys}} = g_i(p) + d_i^{\text{out}}$ with $|d_i^{\text{out}}| \leq b_i$, and the delivered numerical output obeys $|z_i^{\text{del}} - g_i(\hat{p})| \leq \rho_i$. Then

$$|z_i^{\text{phys}} - z_i^{\text{del}}| \leq B_i := \sum_j L_{ij} r_j + b_i + \rho_i. \quad (70)$$

This triangle-inequality certificate separates input/calibration uncertainty, output-model discrepancy and representation/evaluation error. Effects already included in p are not counted again in b_i . The R11 exact expression can be retained with no numerical rendering error in its denotation; a finite rendering needs its own ρ_i certificate. Existing binary64 execution is not assigned $\rho_i = 0$ by this new formalization.

For a requested absolute error tolerance $\tau_i \geq 0$, the condition $B_i \leq \tau_i$ is sufficient on the stated domain and acquisition interval. For an allowed physical operating band $[\ell_i, u_i]$, sufficient containment is

$$z_i^{\text{del}} - B_i \geq \ell_i, \quad z_i^{\text{del}} + B_i \leq u_i. \quad (71)$$

Failure to prove this containment does not prove violation: a conservative enclosure may overlap both regions. A wholly disjoint enclosing interval proves exclusion; overlap remains unresolved. This is the general form of R12's bounded bridge and timing decisions. It certifies the declared quantity and interval, not every later state or a different physical application.

5.6 Minimal specialization record

Each specialization binds only: the measurand and conditions; the complete measurement/calibration expressions and actual source records; the unit, frame and acquisition-time realizations; the joint uncertainty or statistical model; and the requested output tolerance with its scoped evidence. The certificate binds those same dependency versions. Changing a calibration, clock, model basis, duration or tolerance requires reevaluating the affected claim, not changing the one-bit packing format.

These are metadata and template/dependency obligations under XOP-R1. Set images, universal bounds and physical evidence are not secretly new executable opcodes. A backend may return UNKNOWN, MISSING_INPUT or RESOURCE_LIMIT when it cannot establish an obligation. The bridge, thermal and optical expressions remain useful before that backend exists.

Metrological traceability is a property of a measurement result supported by its calibration chain and uncertainty; source hashing and exact arithmetic preserve provenance but do not supply that chain. Fitness for a particular use additionally depends on the required uncertainty or tolerance. [ME7: NIST traceability policy] The additions here make that distinction actionable through (70), rather than adding unrelated domain rules.

6 What this addition changes in the formalization

The additions supply missing physical connections and one explicit finite update. They are applied where their hypotheses hold; adding a law name to an unrelated kernel supplies no additional guarantee.

Interface	Strengthened definition	Consequence
Physical specialization	Named balances, closure, reduction, evolution and observation	An application exposes exactly which physics its claims use
Linear mechanical step	Midpoint solve and exact work/dissipation identity	Fixed passive model has no extra numerical loss in that storage balance
Coupled energy	Signed power ports and nonduplicating stores	Dissipation or absorbed energy is transferred once
Thermoelastic material	Free-energy stress/entropy and reciprocal heat coupling	Deformation and temperature no longer use unrelated balances
Optical material	Maxwell and polarization dynamics with energy accounting	Transfer, stored excitation and loss share a declared model
Measurement and calibration	SI constants, acquisition and joint uncertainty	Exact nominal computation is tied to the actual measured quantity
Output tolerance	Bounded-domain propagation plus model and output budgets	A claim is tied to a specific output and admitted conditions

6.1 A specialization now has an explicit completion criterion

For a chosen application, completion of the mathematical profile means that the state and inputs are declared, equations and domains close, relevant boundary exchanges are accounted for, the finite evolution is named, and the requested output has a defined meaning. A derived invariant must cite the particular model and update for which it holds. A physical tolerance additionally needs justified input and model bounds for that application.

This is a useful architecture for specialization because a new kernel can reuse exact arithmetic, words, delayed state, units and source bindings while importing only the required physical equations. The worked finite mechanical profile demonstrates that separation: its rational solve and energy identity need no general electromagnetic solver. An optical-material specialization, in contrast, needs the selected electromagnetic and thermal closure.

6.2 Implementation obligations remain narrow and visible

No new XOP runtime, automatic compiler, proof checker, material calibration or physical executor is claimed. The discrete midpoint equations provide a complete mathematical step on their domain, but a program must still implement the exact field solve, state commit and output

contract. The new material models remain conditional on their parameters, initial/boundary data and chosen finite representation. Their physical applicability needs corresponding evidence.

The profile does not replace arbitrary continuum fields by a finite state without a declared reduction. It does not equate energy balance with correct trajectory, make a physical clock exact by encoding its ticks, or restore information lost before acquisition. It also changes no earlier satellite accuracy result and adds no general quantum or gravitational theory.

6.3 Traceable source and derivation records

Record	Relative release path
Law-profile and finite step	formal/PHYSICS_PROFILE.md
Mechanical/thermal sources	formal/ENERGY_AND_MECHANICS.md
Electromagnetic sources	formal/ELECTROMAGNETIC_LAWS.md
Measurement/constant sources	formal/MEASUREMENT_AND_CONSTANTS.md
Decisions and proof coverage	formal/DECISIONS_AND_PROOFS.md
Preserved parent identity	source/PARENT_BINDING.json
Document and logical review	review/
Distribution integrity	source/release_manifest.json

The three domain records link to the inspected primary and official sources used for their laws and conventions. The document's equations and proofs state the adopted scope explicitly. Parent files remain unchanged; the distribution contains this focused supplement and its editable sources.